
EGO-Evolution: Constraint-Aware Evolutionary AutoML with LLM Hints and Verifiable Provenance for Tabular Binary Classification

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Abstract

1 We present **EGO-Evolution**, a compact Track-1 pipeline that reframes model se-
2 lection as a *constraint-aware* multi-objective search problem: maximize predic-
3 tive performance (AUC) while satisfying practical constraints (e.g., training time
4 and feature budget). We encode candidate pipelines as genomes combining (i) a
5 feature selection hyperparameter k (SelectKBest) and (ii) a classifier family (Lo-
6 gistic Regression, Random Forest, GBDT, SVM) with tunable hyperparameters.
7 We then run an evolutionary search that optionally uses lightweight LLM sug-
8 gestions for mutation, and we output verifiable artifacts (configs, seeds, metrics,
9 traces) for reproducibility. Across three domains (Semi, Bio, Manuf), we report
10 mean $\pm$ std results over five random seeds and conduct ablations removing LLM
11 hints, constraints, verification, and Pareto selection. Our results highlight that (i)
12 the pipeline is competitive on Bio and Manuf, and (ii) on Semi, stronger classi-
13 cal baselines can dominate, motivating more careful constraint/verification design
14 and dataset-specific handling.

15 1 Introduction

16 Practical machine learning rarely optimizes a single objective. In many real settings, a model
17 must achieve high predictive performance while complying with operational constraints such as
18 limited training time, limited feature budgets, or reproducibility requirements. Track-1 of the AI
19 Co-Scientist Challenge Korea motivates solutions that are not only accurate but also *verifiable* and
20 *reproducible* under constraints.

21 We propose **EGO-Evolution**: an evolutionary AutoML pipeline that searches over simple, inter-
22 pretable tabular pipelines while explicitly acknowledging constraints. We structure the problem as
23 multi-objective optimization:

$$\max \text{AUC} \quad \text{subject to} \quad \text{constraints (time/features)}.$$

24 Implementation-wise, we track a constraint satisfaction notion and use Pareto-based selection during
25 evolution.

26 Contributions.

- 27 • A minimal, reproducible **constraint-aware evolutionary search** for tabular binary classi-
28 fication combining feature selection and classifier choice.
- 29 • Optional **LLM-guided mutation hints** (kept lightweight and easily disabled) to explore
30 better hyperparameters.

- **Verifiable provenance outputs:** deterministic configs, seeds, trace logs, Pareto dumps, and best-metrics JSON for each run.
- A **multi-domain evaluation** (Semi/Bio/Manuf) with ablations and baseline comparisons (mean $\pm$ std over five seeds).

2 Method

2.1 Genome (search space)

Each candidate pipeline is represented as a genome:

$$g = (\text{model_type}, k, \theta),$$

where k is the number of selected features and θ are model hyperparameters. We consider four model families: Logistic Regression, Random Forest, Gradient Boosting (GBDT), and SVM. Feature selection uses SelectKBest with ANOVA F-score.

2.2 Evolutionary search

We use a simple $(\mu + \lambda)$ style evolution:

1. Initialize a population by sampling random genomes.
2. Evaluate each genome by fitting a pipeline on train and scoring on validation with AUC.
3. Select elites by ranking (AUC, auxiliary objective) and Pareto-front filtering.
4. Generate offspring via crossover and mutation.

2.3 Constraints and multi-objective structuring

Track-1 is framed as “maximize AUC while satisfying constraints”. We operationalize this as a multi-objective view (AUC $\uparrow$, constraint satisfaction $\uparrow$) and use Pareto-style selection. In this submission artifact, we primarily report AUC/ACC, while constraint and training-time are logged in run artifacts for verification.

2.4 LLM-guided mutation (optional)

We optionally query an LLM for mutation suggestions (e.g., propose a better hyperparameter change). This is a *soft* heuristic: the search remains fully functional with LLM disabled. For Track-1 rigor, we also include a wout_llm ablation.

2.5 Verifiable provenance

Each run writes:

- `config.json` (full run configuration)
- `split.json` (data split metadata / seed)
- `mutation_trace.csv` (per-generation best stats)
- `pareto.csv` (Pareto frontier candidates)
- `best_genome.json` and `metrics_best.json` (best model + val/test metrics)

Additionally, suite-level scripts aggregate results to `summary.csv`, `summary_mean_std.csv`, and `summary_table.md`.

3 Experimental setup

3.1 Datasets and domains

We evaluate on three domains:

Experiment	Val AUC ($\uparrow$)	Test AUC ($\uparrow$)	Test Acc ($\uparrow$)
main_ea_llm_cons_verify_pareto	0.704 $\pm$ 0.012	0.694 $\pm$ 0.014	0.654 $\pm$ 0.021
wout_llm	0.693 $\pm$ 0.012	0.677 $\pm$ 0.016	0.650 $\pm$ 0.023
wout_constraints	0.699 $\pm$ 0.012	0.689 $\pm$ 0.014	0.653 $\pm$ 0.023
wout_verification	0.720 $\pm$ 0.011	0.719 $\pm$ 0.013	0.676 $\pm$ 0.018
single_objective_auc_only	0.707 $\pm$ 0.012	0.697 $\pm$ 0.014	0.659 $\pm$ 0.021
random_search	0.713 $\pm$ 0.012	0.703 $\pm$ 0.014	0.663 $\pm$ 0.020
baseline_gbd	0.733 $\pm$ 0.011	0.726 $\pm$ 0.013	0.689 $\pm$ 0.017

Table 1: Results on the SEMI domain (mean $\pm$ std over 5 seeds).

Experiment	Val AUC ($\uparrow$)	Test AUC ($\uparrow$)	Test Acc ($\uparrow$)
main_ea_llm_cons_verify_pareto	0.997 $\pm$ 0.002	0.994 $\pm$ 0.002	0.973 $\pm$ 0.010
wout_llm	0.997 $\pm$ 0.002	0.993 $\pm$ 0.002	0.972 $\pm$ 0.011
wout_constraints	0.997 $\pm$ 0.002	0.993 $\pm$ 0.002	0.972 $\pm$ 0.011
wout_verification	0.997 $\pm$ 0.002	0.993 $\pm$ 0.002	0.972 $\pm$ 0.011
single_objective_auc_only	0.997 $\pm$ 0.002	0.994 $\pm$ 0.002	0.973 $\pm$ 0.010
random_search	0.996 $\pm$ 0.002	0.992 $\pm$ 0.002	0.971 $\pm$ 0.012
baseline_svm	0.997 $\pm$ 0.002	0.995 $\pm$ 0.002	0.974 $\pm$ 0.010

Table 2: Results on the BIO domain (mean $\pm$ std over 5 seeds).

- **Semi**: semiconductor-like tabular dataset (OpenML SECOM lineage).
- **Bio**: binary classification dataset (sklearn breast cancer fallback for offline stability).
- **Manuf**: manufacturing tabular dataset (OpenML AI4I 2020).

3.2 Metrics

We report validation and test AUC and test accuracy (ACC), aggregated as mean $\pm$ std over 5 seeds. AUC is the primary objective.

3.3 Baselines and ablations

Baselines. We compare against fixed pipelines using the same feature selection + a chosen model family (e.g., baseline_gbd, baseline_svm).

Ablations. We remove one component at a time: wout_llm, wout_constraints, wout_verification, and additionally compare with random_search and single_objective_auc_only.

4 Results

Tables 1–3 summarize performance across the three domains.

4.1 Key observations

Bio and Manuf are near-saturated. All methods achieve very high AUC (≈ 0.99), suggesting these domains are easy for the considered model class. In such regimes, ablations have minimal observable impact.

Semi highlights the importance of dataset-specific robustness. On Semi, the best baseline (GBDT) outperforms the proposed configuration in test AUC. This indicates that while constraint-aware evolutionary selection is principled, careful handling of dataset quirks (e.g., inactive versions, target metadata issues, missing labels, feature constancy) and search dynamics is critical.

Experiment	Val AUC ($\uparrow$)	Test AUC ($\uparrow$)	Test Acc ($\uparrow$)
main_ea_llm_cons_verify_pareto	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
wout_llm	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
wout_constraints	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
wout_verification	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
single_objective_auc_only	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
random_search	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004
baseline_rf	0.998 $\pm$ 0.001	0.998 $\pm$ 0.001	0.985 $\pm$ 0.004

Table 3: Results on the MANUF domain (mean $\pm$ std over 5 seeds).

Verification can change behavior on Semi. On Semi, wout_verification achieves higher AUC than the full version. This suggests that the verification logic (or its interaction with selection and caching) can influence the explored regions of the search space; this is a concrete direction for improving the “provenance-first” design without harming accuracy.

5 Limitations

- **Constraint metric reporting:** although constraint/time information is logged per run, the current aggregated tables focus on AUC/ACC. Future versions should include aggregated constraint satisfaction and training-time distributions.
- **Semi domain instability:** OpenML SECOM lineage can be inactive or have missing target metadata depending on version/caching; robust fallbacks are necessary and can affect comparability.
- **Simple model class:** the search space is intentionally compact (classical ML + K-best). While this improves interpretability and speed, it may miss gains possible with richer feature engineering or calibrated probabilistic models.

6 Conclusion

We introduced EGO-Evolution, a minimal constraint-aware evolutionary AutoML pipeline with optional LLM-guided mutations and verifiable provenance outputs. Across three domains, the method is competitive on Bio and Manuf and reveals important stability considerations on Semi. These results support the Track-1 goal of building not only accurate but also reproducible and verifiable AI workflows, while motivating future work on stronger constraint reporting and dataset-specific robustness.

A Reproducibility artifacts

Each run saves config.json, best_genome.json, metrics_best.json, mutation_trace.csv, and pareto.csv. Suite-level scripts aggregate them into summary.csv, summary_mean_std.csv, and summary_table.md for each domain.

AI Co-Scientist Challenge Korea paper checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]. **Justification:** The abstract/introduction describe a constraint-aware evolutionary search with optional LLM hints and verifiable artifacts, matching the implemented pipeline and reported results.

122 **2. Limitations**

123 **Question:** Does the paper discuss the limitations of the work performed by the authors?

124 **Answer:** [Yes]. **Justification:** See the “Limitations” section on constraint aggregation, dataset
125 instability (Semi), and compact search space.

126 **3. Theory Assumptions and Proofs**

127 **Question:** For each theoretical result, does the paper provide the full set of assumptions and a
128 complete proof?

129 **Answer:** [N/A]. **Justification:** This work is empirical and system-oriented; it does not propose new
130 theorems.

131 **4. Experimental Result Reproducibility**

132 **Question:** Does the paper fully disclose all the information needed to reproduce the main experi-
133 mental results?

134 **Answer:** [Yes]. **Justification:** We disclose the pipeline components, search procedure, metrics,
135 seeds, and the exact output artifacts (configs, traces, metrics JSON) needed to replicate runs.

136 **5. Open access to data and code**

137 **Question:** Does the paper provide open access to the data and code, with sufficient instructions to
138 reproduce the results?

139 **Answer:** [Yes]. **Justification:** The submission package includes runnable scripts and suite com-
140 mands; datasets are standard OpenML/sklearn sources described in the Experimental setup.

141 **6. Experimental Setting/Details**

142 **Question:** Does the paper specify all training and test details necessary to understand the results?

143 **Answer:** [Yes]. **Justification:** We specify model families, feature selection, metrics, multi-seed
144 aggregation, and ablation/baseline definitions.

145 **7. Experiment Statistical Significance**

146 **Question:** Does the paper report error bars or other statistical significance information?

147 **Answer:** [Yes]. **Justification:** We report $\text{mean} \pm \text{std}$ over five seeds in Tables 1–3.

148 **8. Experiments Compute Resources**

149 **Question:** Does the paper provide sufficient information on compute resources needed?

150 **Answer:** [No]. **Justification:** We log per-run elapsed seconds, but we did not fully standardize and
151 report CPU model/RAM details in the paper; this can be added in the final version.

152 **9. Code Of Ethics**

153 **Question:** Does the research conform with the NeurIPS Code of Ethics?

154 **Answer:** [Yes]. **Justification:** We use standard tabular datasets and do not involve human subjects,
155 sensitive attributes, or deployment.

156 **10. Broader Impacts**

157 **Question:** Does the paper discuss potential positive and negative societal impacts?

158 **Answer:** [Yes]. **Justification:** The work supports safer ML practice via reproducibility and con-
159 straint awareness; misuse risk is low due to classical models and no personal data.

160 **11. Safeguards**

161 **Question:** Does the paper describe safeguards for responsible release of high-misuse-risk assets?

162 **Answer:** [N/A]. **Justification:** We do not release high-risk generative models or scraped datasets.

163 **12. Licenses for existing assets**

164 **Question:** Are creators/owners properly credited and licenses respected?

165 **Answer:** [Yes]. **Justification:** Data sources are OpenML and sklearn datasets; code relies on
166 standard open-source Python libraries (scikit-learn, numpy, pandas).

167 **13. New Assets**

168 **Question:** Are new assets introduced well documented?

169 **Answer:** [Yes]. **Justification:** The submission includes scripts and structured outputs (configs,
170 traces, summary CSVs) with clear folder conventions.

171 **14. Crowdsourcing and Research with Human Subjects**

172 **Answer:** [N/A]. **Justification:** No crowdsourcing or human-subject data collection is performed.

173 **15. IRB Approvals**

174 **Answer:** [N/A]. **Justification:** No human-subject research is conducted.

175 **References**

176 [1] Vanschoren, J., et al. OpenML: networked science in machine learning. *SIGKDD Explorations*,
177 2013.

178 [2] Pedregosa, F., et al. Scikit-learn: Machine Learning in Python. *JMLR*, 2011.